Project Planning Artifacts

Included files:

- docs/program.md
- reports/ablation_comparison_table.md
- reports/final_two_week_plan.md

Revised Project Statement and Agent Strategy

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Revised Project Statement

This project studies whether weather exposure improves prediction of mental-health and burnout-related outcomes. The core research question is:

> Do weather variables add predictive value beyond worker, demographic, and workplace variables?

The project evaluates this question with two targets:

1. `sought_treatment`: a binary mental-health survey outcome indicating whether a respondent sought mental-health treatment.
2. `burnout_index`: a continuous PCA-derived burnout score constructed from non-weather workplace and worker-state indicators.

The preferred final target is `burnout_index` because it is closer to the burnout research goal. The treatment target is kept as a comparison because it connects directly to the mental-health survey and the weather/mental-health reference project.

Current Research Design

The project uses a controlled ablation design:

1. Train models without weather variables.
2. Train the same family of models with weather variables added.
3. Compare validation performance.

For `sought_treatment`, the primary metric is weighted F1. For `burnout_index`, the primary metric is R-squared.

Weather is not allowed in the construction of the burnout index. This prevents leakage because weather is later tested as an explanatory feature.

Data Sources

- `data/raw/survey.csv`: mental-health in tech survey.
- `data/raw/sleep_health_dataset.csv`: sleep, stress, work, and health dataset used to construct the PCA burnout index.

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- `data/weather/state_weather.csv`: state-capital weather data downloaded through the weather API pipeline.

Weather Strategy

The project supports two weather sources:

- Open-Meteo historical weather, preferred for survey-period alignment.
- OpenWeatherMap current weather, useful for reproducing the reference notebook style but weaker for historical inference.

Weather is joined at the state level. This is simple and reproducible, but it is also coarse. A future improvement would use ZIP, county, or city-level weather matched to observation dates.

Burnout Index Strategy

The burnout index is built with PCA in `scripts/burnout_index.py`.

Included burnout indicators are non-weather proxies such as:

- stress score
- work hours
- sleep quality
- sleep duration
- feeling rested
- cognitive performance
- sleep latency
- wake episodes
- weekend sleep difference

Protective variables are reversed before PCA so higher index values consistently represent higher burnout risk.

Agent Strategy

The agent pipeline is organized around reproducible scripts rather than notebook-only work:

- `scripts/download_weather.py`: download weather data.
- `scripts/weather_merge.py`: clean survey or employee data and merge weather by state.
- `scripts/burnout_index.py`: construct PCA burnout dimensions.

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- `scripts/run_full_experiments.py`: run all controlled model comparisons and regenerate reports.
- `scripts/evaluate.py`: shared evaluation metrics.
- `scripts/model.py`: shared preprocessing pipeline.

The current full experiment runs 22 model variants per run across four controlled runs:

- treatment without weather
- treatment with weather
- burnout index without weather
- burnout index with weather

The runner appends all runs to `results/historical_experiment_log.csv` so historical results are preserved instead of overwritten.

A focused follow-up pass in `scripts/improve_and_importance.py` tests a smaller set of tuned candidate models and computes permutation importance for the best pipeline in each run. This is used for interpretation, not for changing the controlled experiment design.

Current Findings

Latest controlled experiment:

Target	Non-weather best	Weather best	Interpretation
`sought_treatment` weighted F1	0.7962	0.7963	Weather adds a negligible improvement.
`burnout_index` R-squared	0.7938	0.7958	Weather adds a small improvement.

Focused improvement pass:

Target	Non-weather best	Weather best	Interpretation
`sought_treatment` weighted F1	0.7959	0.7897	The focused candidates did not improve treatment prediction.
`burnout_index` R-squared	0.7944	0.7962	Tuned histogram gradient boosting slightly improves burnout prediction.

The current evidence suggests weather may add some predictive value, but the effect is small under the current state-level design. The strongest predictors are still work, sleep, and health-related variables. Weather

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variables such as wind gust, pressure, and room temperature appear in importance results, but their incremental contribution is limited.

Main Limitation

Weather is currently assigned by state-level capital weather, not by each worker's exact location and date. This limits the strength of any conclusion about weather effects.

Ablation or Comparison Table

Ablation and Comparison Table

This table compares the best non-weather run against the best weather-augmented run for each target.

Controlled Experiment Results

Target	Task	Non-Weather Best Model	Non-Weather Metric	Weather Best Model	Weather Metric	Weather Delta	Result
`sought_treatment`	Classification	HistGradientBoosting	0.7962 weighted F1	Logistic_C0.1	0.7963 weighted F1	+0.0002	Tiny improvement
`burnout_index`	Regression	HistGradientBoostingReg	0.7938 R-squared	HistGradientBoostingReg	0.7958 R-squared	+0.0020	Small improvement

Focused Improvement Results

This follow-up pass tested a smaller set of tuned logistic, histogram-gradient-boosting, ridge, and neural-network candidates while keeping the original feature sets.

Target	Task	Non-Weather Best Model	Non-Weather Metric	Weather Best Model	Weather Metric	Weather Delta	Result
`sought_treatment`	Classification	HistGB_lr003_leaf15_l2	0.7959 weighted F1	Logistic_C0.03_balanced	0.7897 weighted F1	-0.0063	No improvement over the controlled best
`burnout_index`	Regression	HistGBReg_lr002_leaf63_l2	0.7944 R-squared	HistGBReg_lr002_leaf63_l2	0.7962 R-squared	+0.0019	Small improvement

Interpretation

Weather does not dramatically change predictive performance. The weather-augmented controlled runs are slightly better on both targets, but the improvement is small enough that it should be described cautiously. The focused improvement pass confirms that burnout prediction can be improved slightly with tuned histogram gradient boosting, while treatment prediction does not improve beyond the original controlled best.

The stronger final claim is:

> Weather variables provide weak incremental predictive value in the current validation experiments, but the effect is small and limited by coarse weather matching and by stronger work, sleep, and health predictors.

Ablation or Comparison Table continued

Current Best Runs

Run	Best Model	Primary Metric	Features	Validation Rows
`treatment_non_weather`	HistGradientBoosting	0.7962	25	147
`treatment_weather_augmented`	Logistic_C0.1	0.7963	44	147
`burnout_index_non_weather`	HistGradientBoostingReg	0.7938	21	20000
`burnout_index_weather_augmented`	HistGradientBoostingReg	0.7958	23	20000

Current Best Focused Burnout Runs

Run	Best Model	Primary Metric	Interpretation
`burnout_index_non_weather`	HistGBReg_lr002_leaf63_l2	0.7944	Best non-weather focused model
`burnout_index_weather_augmented`	HistGBReg_lr002_leaf63_l2	0.7962	Best weather-focused model

Recommended Next Ablations

- Use historical weather aligned to the survey year instead of current OpenWeatherMap weather.
- Compare state-level weather against region-only features to test whether weather is adding information beyond geography.
- Remove location fields such as `state_code` when testing weather, because location can partially absorb weather effects.
- Add repeated random splits or cross-validation to test whether the small weather improvement is stable.

Final Two-Week Plan

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# Final Two-Week Plan

## Week 1: Lock the Research Design

### Day 1

- Finalize the research question and target definitions.
- Use `burnout_index` as the main target.
- Keep `sought_treatment` as a secondary mental-health comparison target.

### Day 2

- Use the historical weather source for the main result.
- Keep OpenWeatherMap current weather as a robustness/reference comparison only.
- Confirm the weather variables included in the final model.

### Day 3

- Run the full experiment suite:
  - treatment without weather
  - treatment with weather
  - burnout index without weather
  - burnout index with weather
- Confirm `results/historical_experiment_log.csv` is updated.

### Day 4

- Add robustness checks:
  - compare weather vs climate-region-only features
  - compare variable importance for weather and non-weather models
  - rerun with one alternate random seed if time allows

### Day 5

- Freeze the best result tables.
- Export final plots.
- Write the results interpretation.
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Final Two-Week Plan continued

Week 2: Final Report and Presentation

Day 6

- Draft methods section:
  - data sources
  - PCA burnout index
  - weather merge
  - model families
  - validation design

Day 7

- Draft results section:
  - ablation table
  - best model comparison
  - weather vs non-weather interpretation
  - variable-importance interpretation

Day 8

- Draft limitations section:
  - state-level weather is coarse
  - weather effects may be indirect and absorbed by sleep, work, and health variables
  - treatment is mental-health related but not identical to burnout
  - predictive improvement is small

Day 9

- Build final presentation slides.
- Include:
  - project motivation
  - pipeline diagram
  - PCA index design
  - ablation table
  - final conclusion

Day 10

Final Two-Week Plan continued

- Final cleanup:
  - verify existing `python scripts/run_full_experiments.py` outputs unless results changed
  - verify report files
  - proofread README and project statement
  - prepare final submission

Final Deliverables

- Revised project statement: `docs/program.md`
- Ablation table: `reports/ablation_comparison_table.md`
- Full result matrix: `reports/experiment_result_matrix.csv`
- Historical log: `results/historical_experiment_log.csv`
- Final report bundle: `reports/complete_experiment_log_bundle.md`
- Final two-week plan: `reports/final_two_week_plan.md`
- Variable importance report: `reports/variable_importance.md`